

Spheropolygons and Spheropolytopes for the simulation of soils

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Dedicada muy especialmente a
Andrea. He sido culpable de
algunas de tus lágrimas y me
has dado mas de lo que merezco,
pero ahora vendrán tiempos mas
felices, lo prometo.

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CONFERENCE AND JOURNAL PAPERS RELATED TO THIS THESIS

1. Computational study of the triggering factors of a landslide. Sergio Andres Galindo Torres, Jose Daniel Muñoz Castaño and Arcesio Lizcano. Presented in GKK 08 Geomechanics Colloquium Karlsruhe 2008.
2. Micromechanical aspects of granular ratcheting. **Fernando Alonso Marroquin**, Sergio Andres Galindo Torres and Yucang Wang. Presented in Powders and Grains 2009 Golden Colorado 13-17 July.
3. Effect of the heating of the intergranular water on the softening of a shear band. Sergio Andres Galindo Torres, Jose Daniel Muñoz Castaño and Arcesio Lizcano . Presented in Powders and Grains 2009 Golden Colorado 13-17 July.
4. New perspectives for Discrete Element Modeling: Merging Computational geometry and molecular dynamics. **Fernando Alonso Marroquin**, Sergio Andres Galindo Torres, Antoinette Tordesillas and Yucang Wang. Presented in Powders and Grains 2009 Golden Colorado 13-17 July.
5. Molecular dynamics simulations of complex shaped particles using Minkowski operators on Voronoi tessellations. Sergio Andres Galindo Torres, **Fernando Alonso Marroquin** and Jose Daniel Muñoz Castaño. Submitted to Physical Review E.
6. A micromechanical study on the softening due to heat dissipation inside a granular shear-band. Sergio Andres Galindo Torres and Jose Daniel Muñoz Castaño. In preparation.
7. Molecular Dynamics simulation of complex particles in 3D and the study of friction due to non-convexity. Sergio Andres Galindo Torres, **Fernando Alonso Marroquin**, Yucang Wang, Dorival Pedroso and Jose Daniel Muñoz Castaño. Physical Review E 79, 060301(R) (2009)
8. Bottlenecks in granular flow: When does an obstacle increase the flow rate in an hourglass?. **Fernando Alonso Marroquin**, Imran Azeezullah, Sergio Andres Galindo Torres and Louise Olsen. Submitted to Physical Review Letters.

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CHAPTER 1

Introduction

The study of soils, and in general of granular materials, has attracted the attention of civil engineers who have to deal with settlements of buildings, earth pressure against retaining walls, stability of slopes and embankments etc. Recently, the number of physicist interested in the subject has increased dramatically. Granular materials are considered a new state of matter with its own properties that differentiated it from the liquid, gaseous and solid states. It is a typical example of what is called a *complex system*: the collective macroscopic behavior of the system cannot be easily predicted from the microscopic non-linear interactions between its constitutive elements. Moreover continuous approaches that have been used by engineers are limited since granular ensembles are by nature discrete systems. A novel trend that is becoming popular is to study the soils as statistical ensembles rather than continuous bodies.

There are two paradigms to describe granular soils. In the macromechanical paradigm, proper of geotechnical engineers, the soil is described by constitutive equations, i.e. by tensorial relationships connecting stress and strain, which, together with the tensorial differential equations of movement, define the system dynamics. There are many different models for these constitutive equations (hypoplastic, viscoelastic, etc.), and each model is especially adequate for the model of a specific kind of soil. The parameters that characterize the soil in each model are obtained from simple laboratory tests. These equations, when numerically solved by finite

differences or finite element methods, are successful to predict the macroscopic evolution of the soil. Nevertheless, and despite of the great developments of the last four decades, there are a large set of phenomena that cannot be properly modeled by this kind of models. Examples are the localization of strain under shear stress and the unbounded incremental accumulation of strain under cyclic loads (ratcheting). The macroscopic constitutive equations that intend to describe these phenomena are full of new parameters that are hard to measure and to explain from a physical basis, and the problem remains open.

The micromechanical paradigm, proper of physicists, looks at the soil as a set of discrete grains that interact with each other by elastic, viscous and frictional non-linear forces. The parameters here are the ones characterizing these forces: Young modulus, static and dynamic Coulomb friction coefficients, restitutive coefficients, etc. These forces, together with the grain shapes and sizes, describe the soil from the microscopic point of view, but they are non-linear and dissipative, and there is very hard to find an analytical prediction of the collective behavior. Although these microscopic interactions are responsible, at the end, of the macroscopic behavior, nobody knows how to relate them with the macromechanical constitutive equations, and this problem has become the *holy grail* of the soil mechanics.

From the physicist's point of view, much of the research has been done by using computer simulations, both 2D and 3D, in the so-called Discrete Element Method (DEM). The granular system is modeled as a set of discrete elements (discs, spheres, polygons, etc.), and the forces between them are computed from the positions and velocities of all grains at each time step, either by assuming that the grain models overlaps (Molecular Dynamics) or by solving the set of equilibrium equations and momentum and energy balances for all grains (Contact Dynamics). Next, the Newton and Euler laws of motion (for translations and rotations, respectively) are numerically integrated to obtain the positions and velocities at the next time step. Nowadays computers allow for tracking the positions and velocities of thousands - and even millions - of grains and the forces among them with such a precision that cannot be obtained from the experiments. Next, microscopic interactions can be added one by one and tested in order to see if they reproduce the macromechanical behavior one is interested in; if this reproduction is ended with success, one has captured the essence of the phenomenon. So, micromechanical simulations are not intended to replace the macroscopic models of geotechnics, but to establish the physical grounds of the macroscopic phenomena and, in consequence, to reveal basic ideas that can be used for the development of more powerful macromechanical models. This is the main goal of the DEM method, and the development of more realistic and easy-computed discrete elements

is a fundamental part of the process to reach such goal.

A great part of the micromechanical studies on soils has been done with two-dimensional models. They are easier to programming, less computer demanding, and able to maintain some of the fundamental aspects of many phenomena of interest. Two main discrete elements are used for 2D simulations: disks [1, 2, 3] and Voronoi polygons [4, 5, 6, 7, 8]. Disks are much simpler to represent, and their interaction laws are much easier to compute, too, but they are not completely realistic in their interactions and some features are often added to reproduce granular behavior. They are also more suitable for statistical studies, like for instance the remarkable work of F. Radjai on disks [95], where a model for the distribution of normal forces between grains was proposed. In contrast, Voronoi polygons are harder to describe, some of their interactions do not allow for an exact computation of the elastic potential energy and their magnitudes are not continuous functions of the relative positions, making the code instable; but they account for the irregular shape of real grains and achieve a more realistic behavior with less particles than discs. With Voronoi polygons, Alonso [47] reported the appearance of the well known phenomenon of ratcheting, first found in experiments, and other researchers have been able to reproduce breaking and fractures, but their complex shape has avoided, for instance, to test the McNamara's suggestion [93] that this ratcheting is just an error in the widely used Cundall-Strack model [1] for friction. Summarizing, neither disks nor polygons contains all desired features for a discrete element method.

Despite of the power of two-dimensional models, there are phenomena that ask for a three-dimensional approach. In fact, Poisson's ratios change a lot from 2-D to 3-D, and the percolation of chain forces is completely different. Moreover, both numerical and laboratory tests suggest that there are some significant differences regarding macroscopic friction and wing crack extensions between the 2-D and the 3-D cases [9, 10]. Nevertheless, it is more difficult to deal with finite particle rotations in 3-D, increasing computer requirements, and the evolution of three-dimensional discrete elements has been much slower than in 2-D. Of course, the first 3-D simulations used spheres as discrete elements [11], and much of the present studies are still done with them [12], but many hopes are set in the use of polyhedra. An example of these models is the package LMGC90, developed by Dubois and Jean [13], which uses digitalized images of real grains and contact dynamics to compute the forces. The main problems here are again the difficulty to define elastic forces between polyhedra with no discontinuities and allowing for an exact computation of the elastic energy.

In a recent work, Pournin and Liebling [15] have introduced a new discrete element, called *spheropolyhedra* that bases on the Minkowski operations of erosion and dilation employed in graphics processing. The idea is to obtain rounded particles, with interactions that can account for the elastic energy at the contacts. The contact law, however, is unnecessarily complex, and is only suitable for convex shapes. Thus, very few results have been obtained so far to validate its utility, and only three different shapes have been modeled (see Fig. 5.5). Later on, Alonso [14] developed a new kind of discrete elements, called *spheropolygons*, with interactions between vertex and sides that are continuous on the relative moves of the grains and analytically defined, with the additional advantage that they allow for an exact computation of the elastic energy. Since the interaction is multicontact among grains, non-convex shapes can be modeled. Nevertheless, they have still not been generated with random shapes, a relevant feature for the simulation of soils.

The objective of the present thesis are, first, to combine the spheropolygons of Alonso with the Voronoi construction in order to create discrete elements that can be used to simulate soils and, second, to extend these ideas to the construction of three-dimensional spheropolyhedra with a simple definition of the elastic forces. After construction, both 2D and 3D models should be checked against the classical conservation laws of physics. Afterwards we test them in the study of soil phenomena like hysteresis, ratcheting and the evolution of full-saturated shear bands, that is where the exact account of the elastic energy is of relevance. This will give to the micromechanical community a set of new and validated tools to investigate the complex behavior of granular soils in studies where shape and energy play a mayor role.

The scope of the thesis is summarized as follows: In chapter 2 we explain in more detail the scope and goals of the present thesis. It is followed by chapter 4 reviewing the foundations of the DEM method that all implementations share. Some important mechanical behaviors of soils are explained in chapter 3 by introducing useful validation examples. Then, the state of the art is explained in detail in chapter 5. Before starting with spheropolgons and spheropolyhedra, chapter 6 uses the simple geometries of disks and spheres to address three interesting problems: The effect of an obstacle to speed up the flow in the bottleneck of an hourglass, the numerical nature of the reported ratcheting in DEM simulations (mentioned above), and the study of the distribution of elastic forces in a sample of monodisperse spheres under isotropic compression. These problems will prove that although the disk and sphere geometry is limited, they are still useful modeling tools.

Next, the issue of spheropolygons is fully addressed. Chapter 7 explains how to construct Minkowski spheropolygons from Voronoi constructions. We start with a precise and detailed description of the implementation procedure and then we go further by testing for both efficiency and conservation laws with simple simulation setups. Afterwards, these constructions are employed in chapter 8 to simulate the biaxial test of soils under both monotonic and cyclic loadings. First, the setting of the critical state and the orientation of the shear bands are investigated. Next, we explore some phenomena where a proper account of the elastic energy is relevant: the appearance of hysteresis loops and ratcheting. Then, chapter 9 faces the simulation of a fully-saturated shear band by coupling our DEM code with a simple thermodynamical model for the intergranular water. The goal is to investigate the vaporization hypothesis of Habib [111] for the triggering of a landslide. This last example illustrates the power of the new discrete element method and opens new interesting research lines.

Thereafter, we extend the model to three-dimensions. Chapter 10 describes how to build spheropolytopes. The implementation features are very similar to its 2D counterpart, but the differences are explained in detail. Chapter 12 performs some simulations with samples of identical spheropolytopes that illustrates the many application possibilities of this element method: The collision between jets of Tetrahedra, which test for energy, linear and angular momentum conservation, the hooper flow of non-convex particles, and the emergence of macroscopic friction from the microscopic interaction between two rough surfaces. Finally, chapter 13 address the simulation of a soil of Minkowski-Voronoi spheropolyhedra undergoing the true triaxial test. We look first for the critical state and compare with the same simulation for spheres, and the role of particle shape on the macroscopic friction and dilatancy angles. The chapter ends by comparing linear elastic forces against a full implementation of the Hertz [124] and Puttock [104] forces among spheres and cylinders. Nature of soil is studied with the simulation of a True Triaxial Test comparing the macroscopic response of different geometries and different contact laws. Finally, chapter 13 summarizes the main conclusions and gives some final discussions and projections of the present work.

CHAPTER 2

The Research Problem

As was mentioned in the last chapter, the use of Voronoi polygons has been a very successful DEM model for the two-dimensional simulation of granular media and soils. The force for these discrete elements depends on the overlapping area between polygons. This produces two main problems.

The first one deals with the efficiency and stability of the method. Two polygons of N sides will require an algorithm of order $O(N^2)$ to compute the overlapping area, because we have to check every side of each polygon to see if there is an overlap and how big it is. The computer first check for possible intersections between sides and, if an intersection is found, proceed to integrate the enclosed area. Obviously, the more complex the polygon, the more CPU time is required to compute the area. The second problem deals with the calculus of the elastic energy. Unfortunately, a force proportional to an area, but not to a distance, cannot be derived from a potential, and the elastic energy cannot be obtained.

The spheropolygons developed by Alonso[14] solves these problems by defining the force between two spheropolygons as a function of distances among them, and therefore, a potential can be easily found. Hence we propose to use this method to investigate soils, especially where computing the elastic energy can be of fundamental relevance for the phenomenon of interest.

The work of Alonso on spheropolygons has not included yet a Voronoi construction for general shapes. Given that such Voronoi construction is a well established method in DEM simulation of soils and geomechanics, our first goal is to combine these two concepts to create Voronoi spheropolygons and reproduce simulations done with the Voronoi method: the incremental response on biaxial tests and the ratcheting under cyclic loads. Moreover, computing the elastic potential energy allows us to check for energy balance in more complex problems, like the hysteresis area under cyclic loads or the heating of a shear band by effect of the dynamic friction between grains. Basically, with this potential energy one hand we can compute the dissipation by cycle, in the first problem, or the proper energy balance, in the second one. This completes the first part of the project.

The second part of this project deals with the extension of these discrete elements to 3-D. Actually, the development of three-dimensional polyhedra with a simple description of forces is a main need for the molecular dynamics simulation of granular media, because Voronoi polygons are very difficult to generalize to three dimensions. Computing the overlapping volume of two polyhedra is an extremely complex task, and the solution will inherit both problems mentioned above. In addition, the implementation of spheropolyhedra by Pournine [16] has an overcomplicated contact law that limits its application to soil micromechanics. In contrast, the definition of elastic forces by Alonso is much simpler, and the basic concepts can be easily extend to 3-D, given rise to a broad spectrum of future applications.

Spheropolyhedra offer a lot of other applications. Simulation of soil grains with realistic shape is just one of them. It can simulate drugs pills, molecules, effects of non convex shapes like friction, toys like tops, etc. Industries dealing with soils, mining, video games and more will benefit from this method.

This is, summarizing, the main scope of the present Phd thesis. We hope it will extend the micromechanical study of soils with a new set of discrete elements that will contribute to develop a broad horizon of possibilities in the simulation of soils.

CHAPTER 3

A Brief Introduction to Soil Mechanics

Soils are the main subject of study of civil engineers who have to deal with settlements of buildings, earth pressure against retaining walls, stability of slopes and embankments, etc. These geotechnical engineers describe the soil as a continuous media and characterize them by means of tensorial constitutive equations, intending to summarize the main phenomenological aspects that are consequence of the discrete nature of the soil. The parameters of these macromechanical models are obtained from elementary essays in the laboratory, which are also used to the systematic study of the very interesting phenomena.

Hereby we summarize some of the basic theoretical models for soil mechanics, describe some of the elementary tests employed to characterize soils that will be simulated in the project, and explain some of the phenomenological behaviors that are challenging for continuous models: hysteresis, ratcheting and the evolution of full-saturated shear bands.

3.1 Constitutive Models and Failure Criteria

A constitutive model is basically a relationship between the measured strain ϵ and the stress σ applied on a continuous body,

$$\epsilon = f(\sigma, \kappa) \quad , \quad (3.1)$$

with κ a set of parameters describing the material. The easiest model is of course linear elasticity, but most aspects of soil behavior cannot be explained by linear responses, as will be evident below. Many different models have been proposed in the last four decades, but, as Kolymbas pointed out [17], they are very hard to reproduce and even to understand by other researchers, and many of the parameters defined by these models are hard to measure or to relate with physical concepts. Hereby we introduce two of them: the elasto-plastic theory of Drucker and Prager and the hypoplastic theory of Gudehus.

3.1.1 The Mohr-Coulomb Criterion

The initial response of a soil to an applied stress is linear elasticity. This only happens if the stress is not high enough to fail the soil.

The Mohr-Coulomb failure criterion[18] considers this elastic regime as the only possible state until the failure is reached,

$$\sigma_t = c + \sigma_n \mu \quad , \quad (3.2)$$

and relates the the normal σ_n and deviatoric stress σ_t in terms of the friction coefficient μ and the cohesion coefficient c .

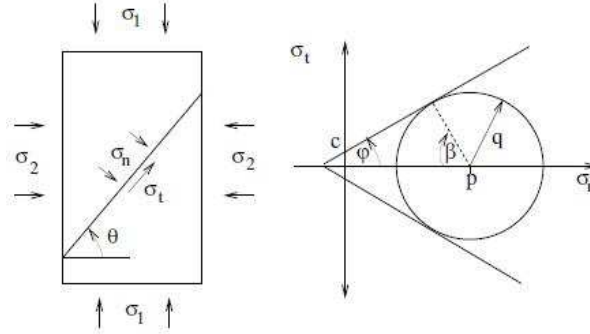


Figure 3.1: A sample is subjected to principal stress and a failure is obtaining with an angle θ from the horizontal (left). The failure condition of Eq.3.2 is represented by a cone in the stress space (right)

Fig3.1 illustrates the basic idea behind this criterion in 2D. The space of possible normal to tangential stresses on all possible orientation planes draws a circle: the Mohr circle, subtended between the values of the principal stresses. This circle has center at $p=(\sigma_1 + \sigma_2)/2$ and radius $q=(\sigma_1 - \sigma_2)/2$. The Coulomb criterion establishes a straight line, with slope μ , for the space of normal to tangential stresses at failure. Thus, if the stresses increase and the Mohr circle grows to touch the Coulomb line, there is a plane where the tangential stress will reach the maximal value given by the Coulomb criterion and, therefore, the material fails. The angle β of the intersection point is related to the angle θ of the failure plane by $2\theta=\pi - \beta$. So, this criterion gives us information about the occurrence and orientation of the failure.

3.1.2 Elasto-plastic theory

When a granular sample undergoes a confining pressure and therefore an axial stress is applied, it displays the typical stress-strain relationship shown in figure3.2. A continuous decrease of the stiffness (i.e. the slope of the stress-strain curve) is observed during loading. If the sample is unloaded, an abrupt increase in the stiffness is observed, leaving an irreversible deformation. Despite of that, one can approximate that the deformation is almost linear and reversible if the stress changes inside some region around σ_A , which is called the *yield point*.